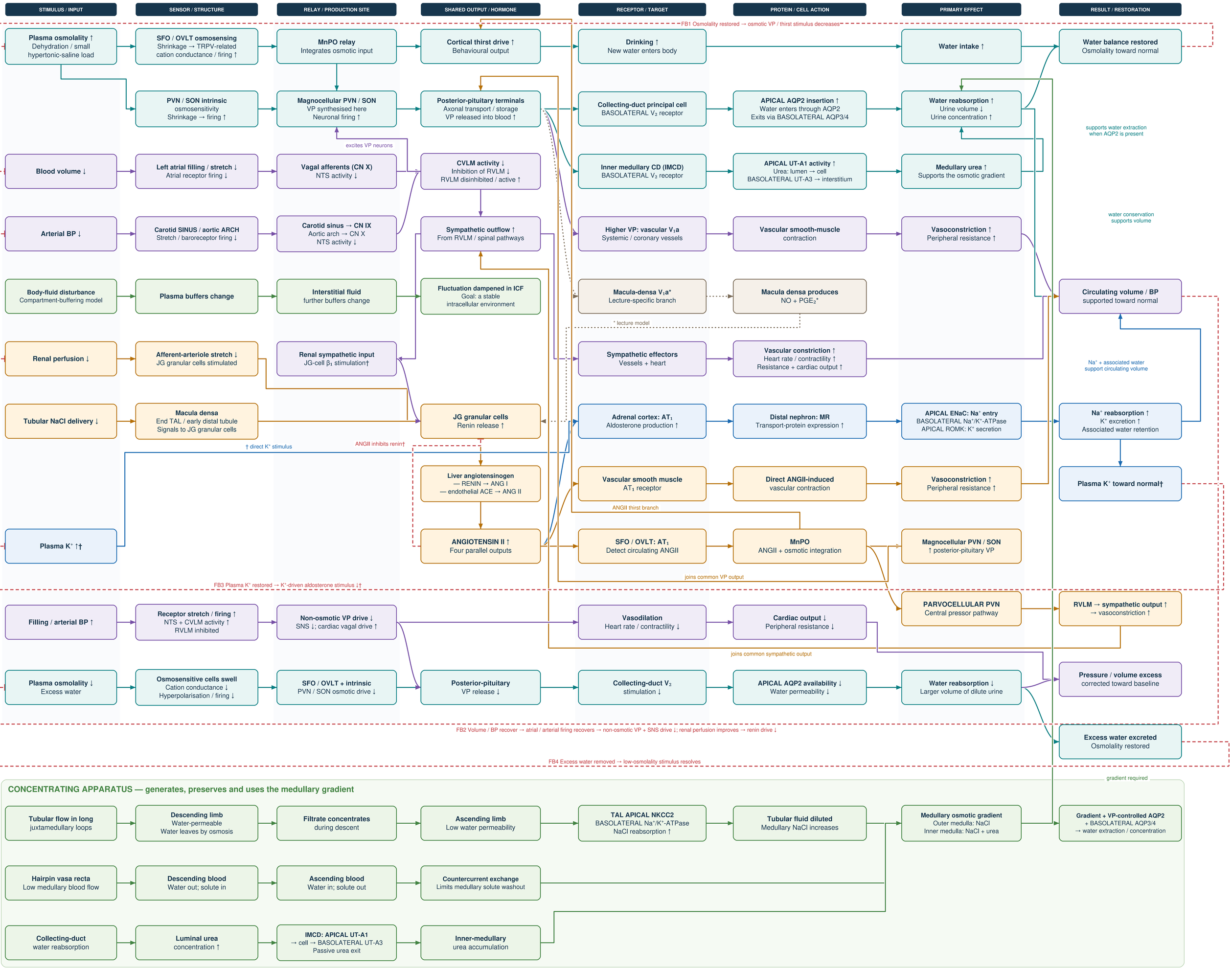
RENAL PHYSIOLOGY — ALL PATHWAYS

STIMULATION ON THE LEFT → EFFECT ON THE RIGHT | Shared nodes are connected; crossings without a dot are not junctions.

Key structures, receptors and transport proteins retained. Intracellular signalling / phosphorylation-site details omitted.



READING KEY

Solid coloured arrows = forward pathways. Red dashed flat-ended arrows = NEGATIVE FEEDBACK. Dashed taupe = lecture-specific VP–macula-densa model.

* Macula-densa V_1a → NO / PGE_2 → renin is the lecture model; net renin also reflects perfusion, NaCl delivery and pressure / volume feedback.

† Supporting physiology added for complete connections: JG β_1 input, direct K^+–aldosterone control, ANGII–renin feedback and distal acid-excretion links.

PVN / SON produce VP; posterior-pituitary terminals release it. SFO / OVLT are sensory sites; PVN / SON also have intrinsic osmosensitivity.

MnPO = median preoptic nucleus; NTS = nucleus of the solitary tract; CVLM inhibits RVLM. M = magnocellular; parvocellular PVN provides the central SNS branch.

Source: Dr J Chu, BBMS2006 (Renal — student version), PDF pp. 3–15, 19–46; relevant classroom cardiovascular / saline material. AQP2 needs an intact medullary gradient.

Same named structures in different rows are the same anatomical sites. Water concentration, total Na^+, ECF volume and arterial BP are related but distinct variables.